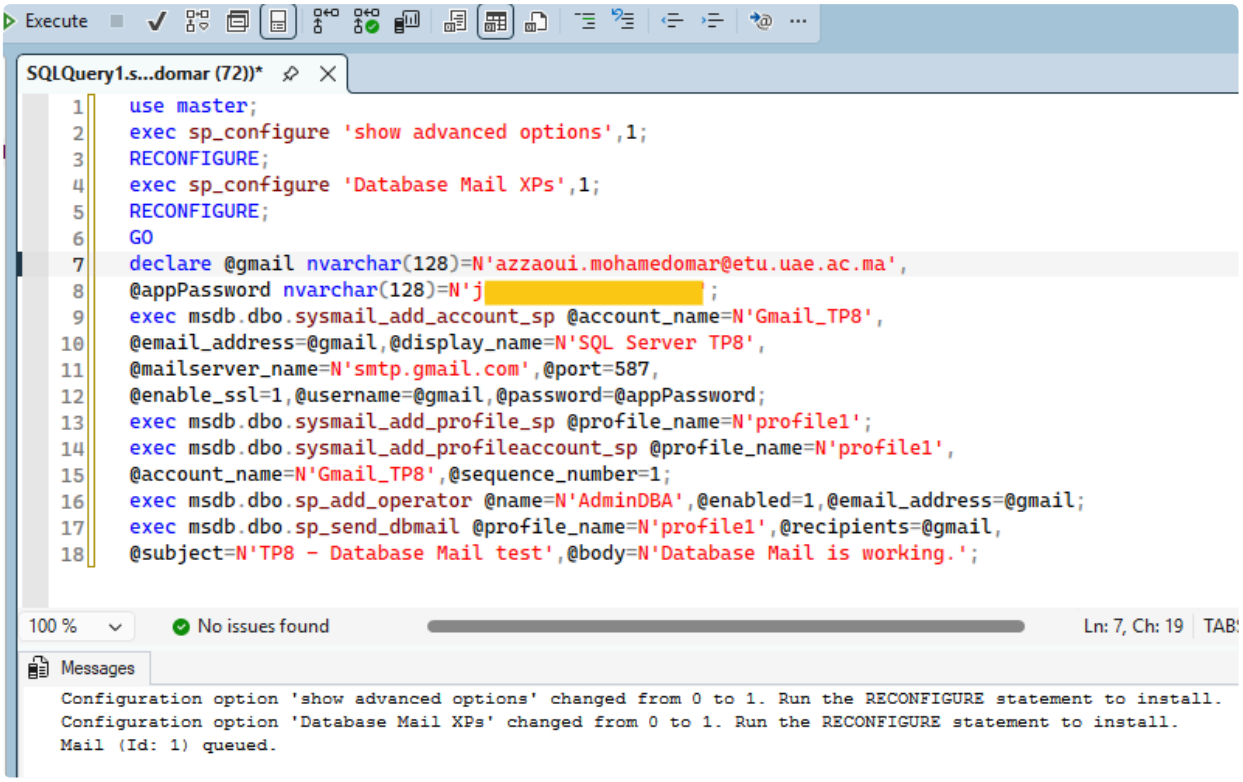
TP8 - Automation and monitoring with SQL Server Agent

Preparation

The lab uses the default SQL Server instance on WIN11. SQL Server Agent is running with automatic startup.

Exercise 1 - Database Mail and operator

Database Mail was configured with an SMTP account, a Google app password and the profile `profile1`.



The test email was received successfully.



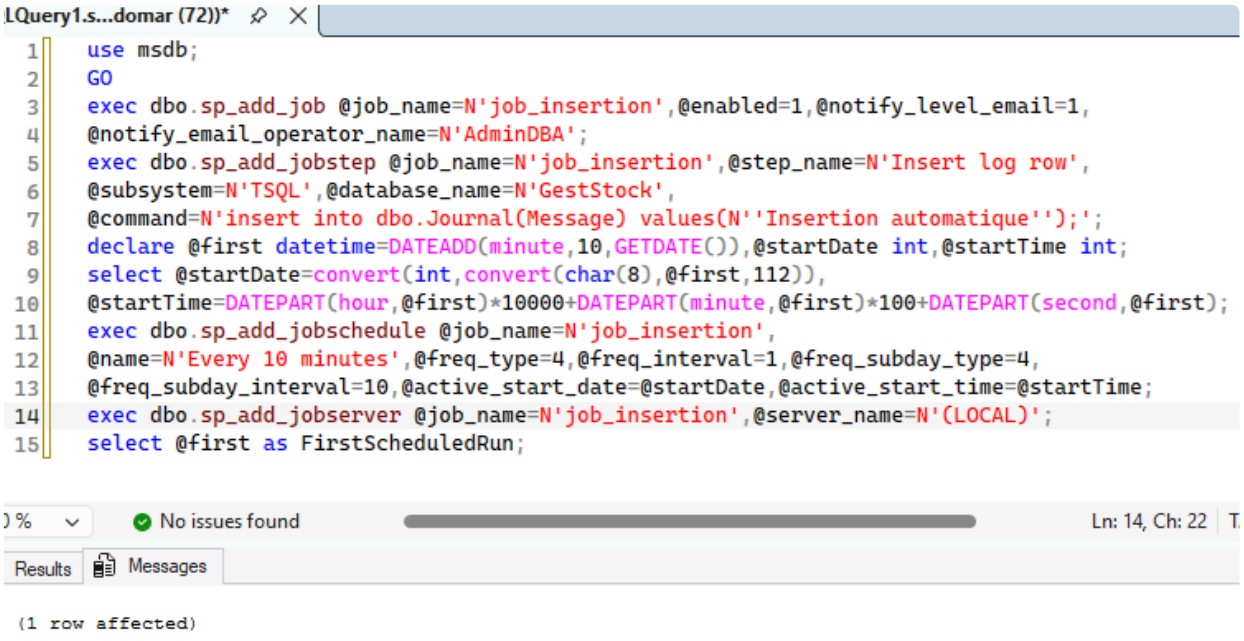
In **SQL Server Agent** → **Properties** → **Alert System**, the mail profile was enabled with **Database Mail** and `profile1`. The Agent was then restarted to apply the profile.

Exercise 2 - Scheduled insertion and notifications

Job creation

The job `job_insertion` inserts `Insertion automatique` into `GestStock.dbo.Journal` every 10 minutes. Initially, `@notify_level_email=1` sends an email to `AdminDBA` after each successful execution.

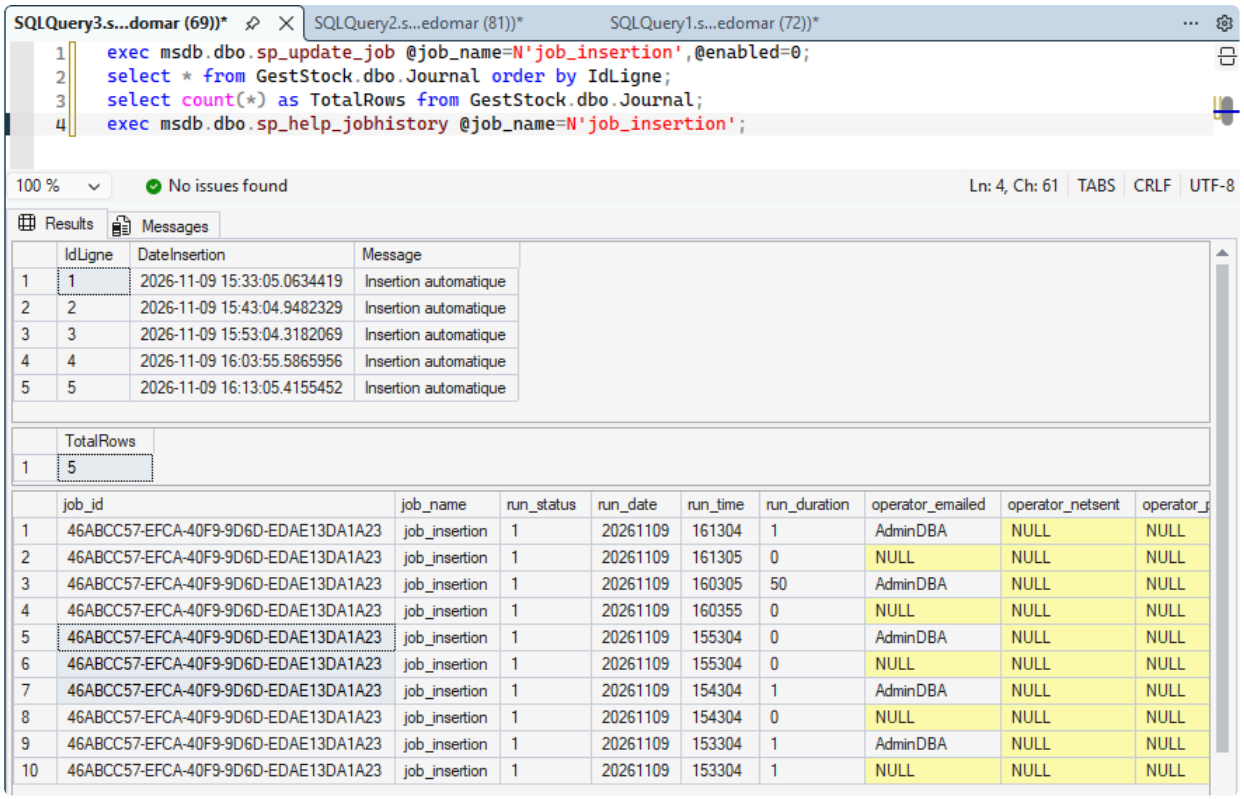
Procedure	Role
`sp_add_job`	Creates the job and configures notifications
`sp_add_jobstep`	Defines its SQL command and target database
`sp_add_jobschedule`	Attaches the 10-minute schedule
`sp_add_jobserver`	Assigns the job to the local instance



Execution results

The first scheduled run was at **15:33:04**, followed by **15:43:04** and **15:53:04**. With the first execution scheduled 10 minutes after creation, the initial 30-minute period produces **3 rows and 3 success emails**, excluding the Database Mail test and alert emails.

The job remained enabled longer. At the final check, it had completed **5 successful executions**, producing **5 rows and 5 success notifications**. It was then paused before the failure test.



```
exec msdb.dbo.sp_update_job @job_name=N'job_insertion',@enabled=0;
select * from GestStock.dbo.Journal order by IdLigne;
select count(*) as TotalRows from GestStock.dbo.Journal;
exec msdb.dbo.sp_help_jobhistory @job_name=N'job_insertion';
```

100 % — No issues found — Ln: 4, Ch: 61 — TABS — CRLF — UTF-8

Results | Messages

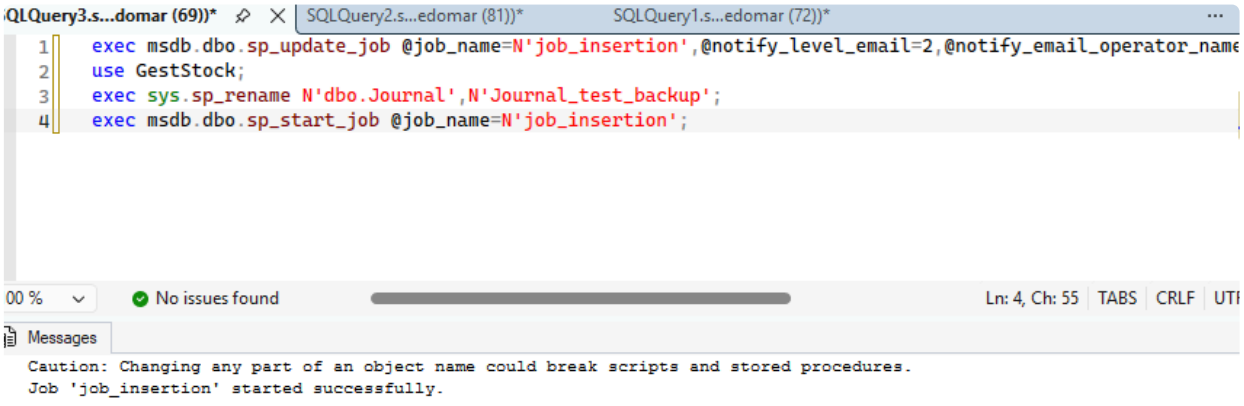
	IdLigne	DateInsertion	Message
1	1	2026-11-09 15:33:05.0634419	Insertion automatique
2	2	2026-11-09 15:43:04.9482329	Insertion automatique
3	3	2026-11-09 15:53:04.3182069	Insertion automatique
4	4	2026-11-09 16:03:55.5865956	Insertion automatique
5	5	2026-11-09 16:13:05.4155452	Insertion automatique

	TotalRows
1	5

	job_id	job_name	run_status	run_date	run_time	run_duration	operator_emailed	operator_netsent	operator_p
1	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	161304	1	AdminDBA	NULL	NULL
2	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	161305	0	NULL	NULL	NULL
3	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	160305	50	AdminDBA	NULL	NULL
4	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	160355	0	NULL	NULL	NULL
5	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	155304	0	AdminDBA	NULL	NULL
6	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	155304	0	NULL	NULL	NULL
7	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	154304	1	AdminDBA	NULL	NULL
8	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	154304	0	NULL	NULL	NULL
9	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	153304	1	AdminDBA	NULL	NULL
10	46ABCC57-EFCA-40F9-9D6D-EDAE13DA1A23	job_insertion	1	20261109	153304	1	NULL	NULL	NULL

Failure-only notifications and failure test

Emailing every success creates unnecessary noise. Successful runs can be checked in job history; notifications should draw attention to failures. The notification level was changed to **2**, then the table was temporarily renamed before a manual execution.



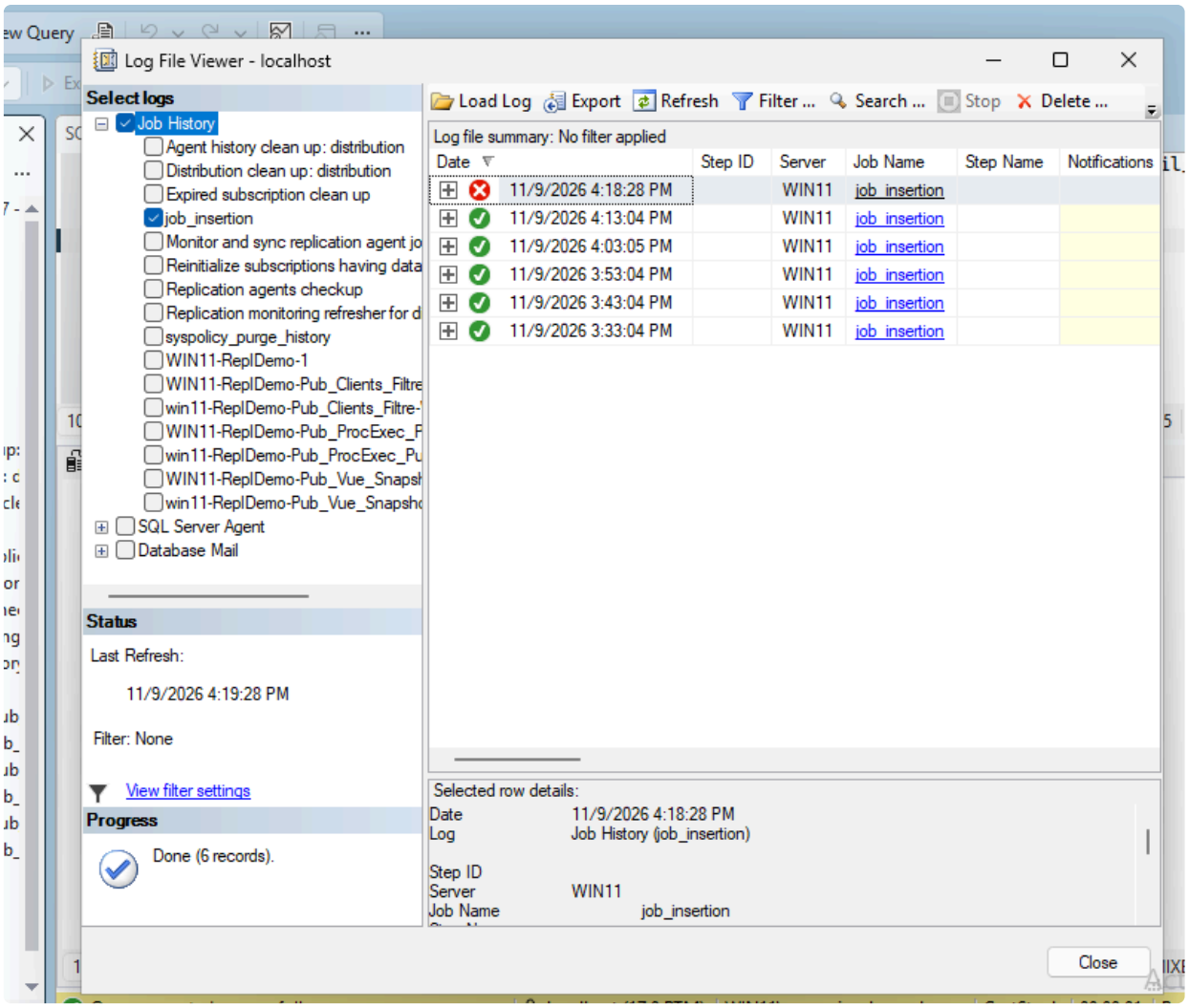
```
exec msdb.dbo.sp_update_job @job_name=N'job_insertion',@notify_level_email=2,@notify_email_operator_name
use GestStock;
exec sys.sp_rename N'dbo.Journal',N'Journal_test_backup';
exec msdb.dbo.sp_start_job @job_name=N'job_insertion';
```

00 % — No issues found — Ln: 4, Ch: 55 — TABS — CRLF

Messages

```
Caution: Changing any part of an object name could break scripts and stored procedures.
Job 'job_insertion' started successfully.
```

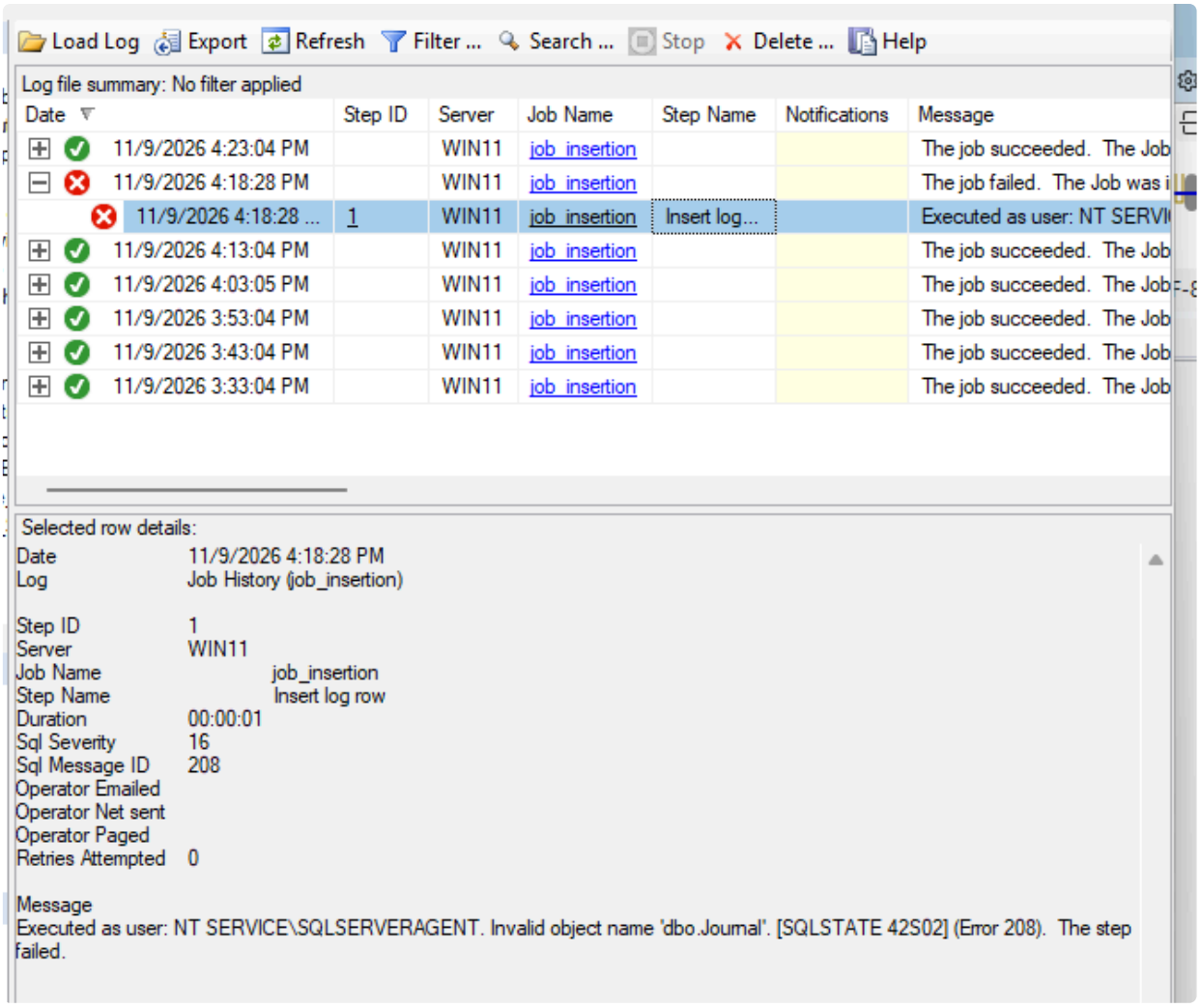
The execution at **16:18:28** failed with **error 208: Invalid object name 'dbo.Journal'**. Job history records the failure, and `AdminDBA` received its notification.





After the failed execution finished, the original table name and scheduling were restored.

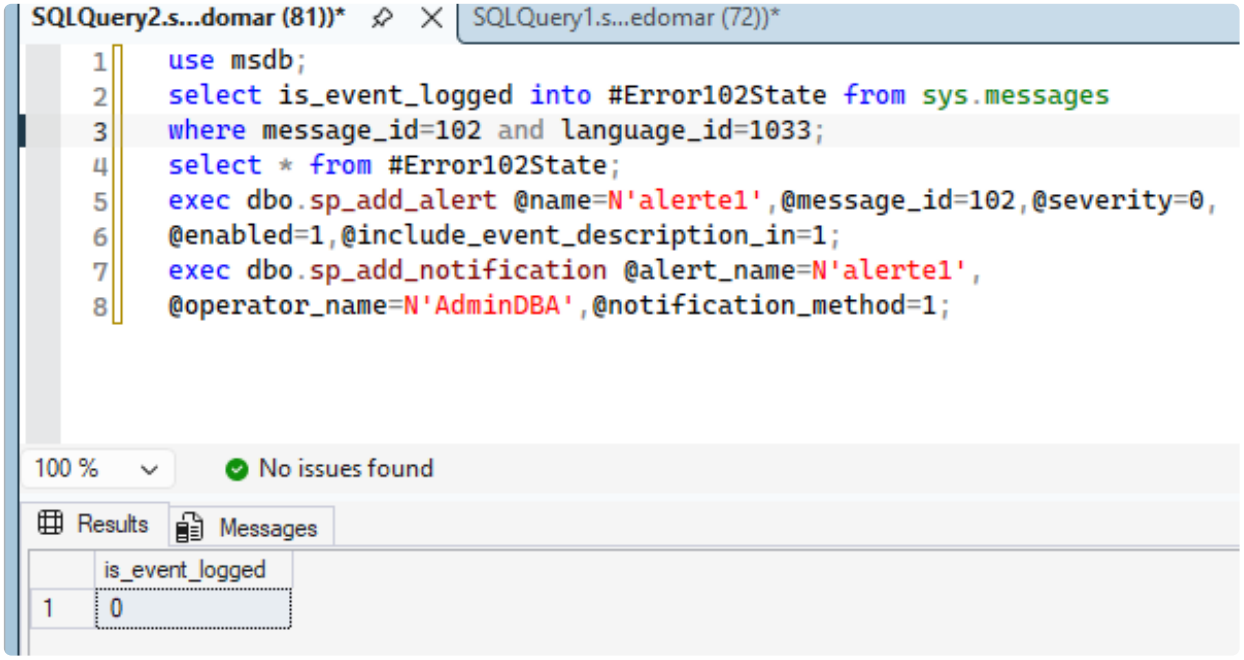
The failed insertion left the count at **5**. A subsequent successful run at **16:23** confirms recovery. Notifications remain set to failure only.



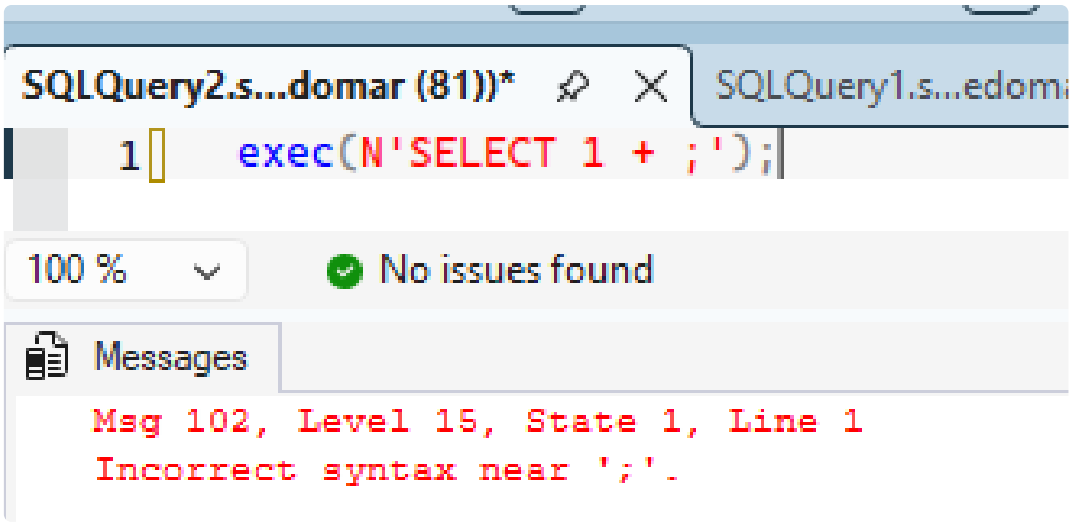
Exercise 3 - Error alerts

Syntax error 102

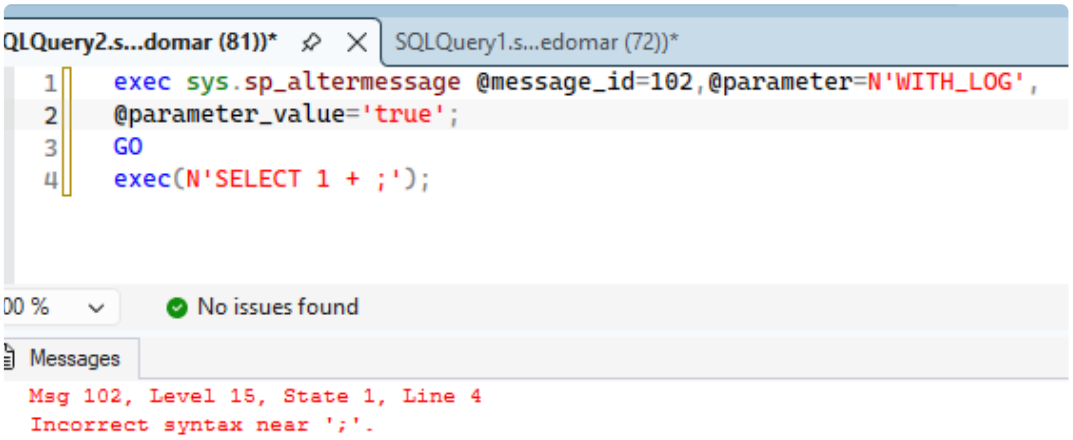
The alert `alerte1` monitors error 102 and emails `AdminDBA`. The original logging setting was saved before testing.



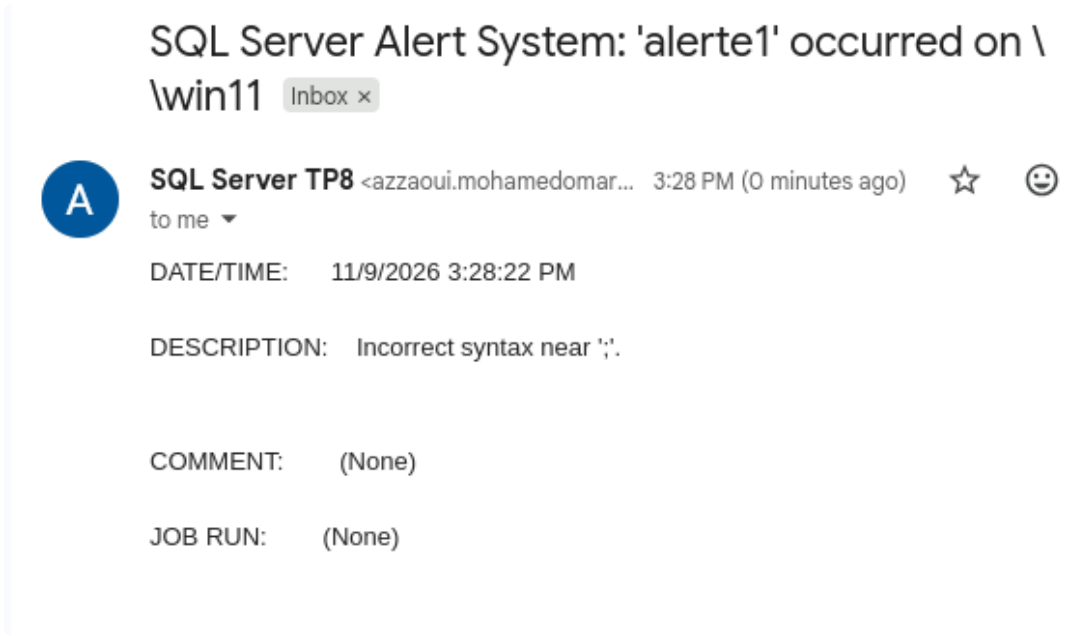
The first test raised error 102.



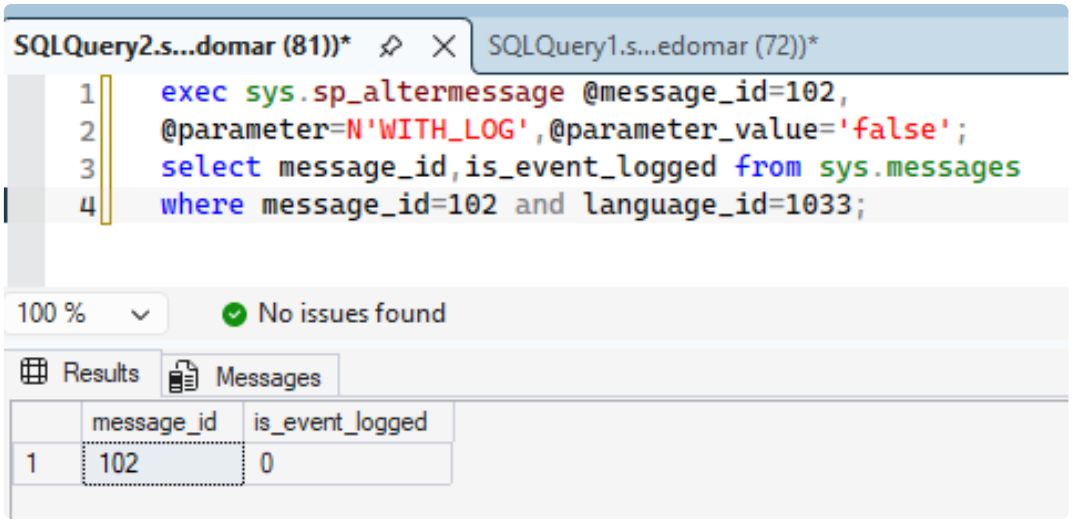
No email was received because `is_event_logged=0`. SQL Server Agent requires the event to be written to the Windows Application log. Enabling logging also writes it to the SQL Server error log.



The repeated test triggered the alert email.

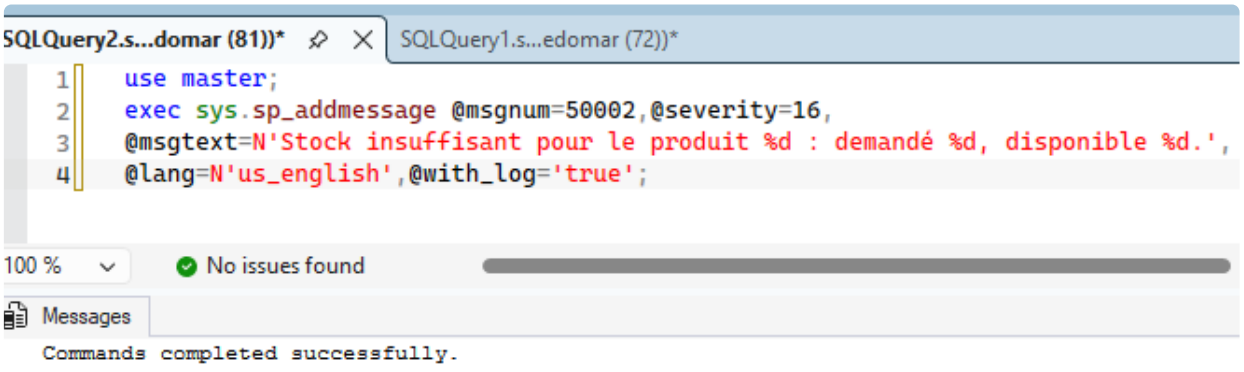


The original setting was restored afterward.



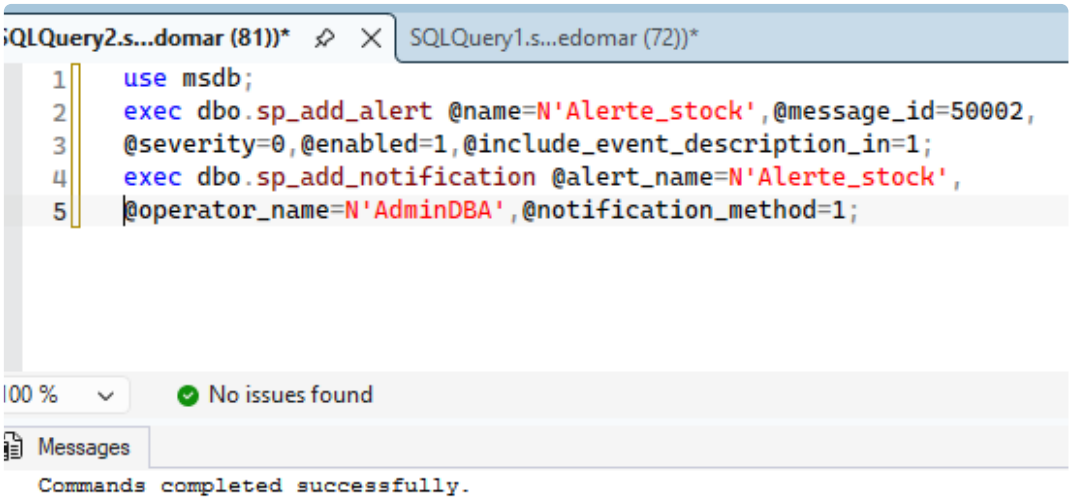
Stock validation and error 50002

The custom error includes the product identifier, requested quantity and available stock. Logging is enabled so it can trigger an Agent alert.



`PasserCommande` rejects invalid quantities and only decreases stock when enough units are available. Insufficient stock raises error 50002.

The alert `Alerte_stock` sends the error description by email. `@include_event_description_in=1` includes it in email notifications, and `@notification_method=1` selects email delivery.



A valid order reduces the keyboard stock from **3 to 2** without raising an alert:

```
QLQuery2.s...domar (81))*   SQLQuery1.s...edomar (72))*
1   use GestStock;
2   exec dbo.PasserCommande @IdProduit=2,@Quantite=1
3   select * from dbo.Produit where IdProduit=2;
```

100 % 1 0

Results Messages

	IdProduit	Libelle	Stock
1	2	Keyboard	2

Requesting 10 units raises error 50002 and triggers `Alerte_stock`. The stock remains **2**.

```
QLQuery2.s...domar (81))*   SQLQuery1.s...edomar (72))*
1   exec dbo.PasserCommande @IdProduit=2,@Quantite=10;
2   GO
3   select * from dbo.Produit where IdProduit=2;
```

100 % 1 0

Results Messages

```
Msg 50002, Level 16, State 1, Procedure dbo.PasserCommande, Line 16 [Batch Start Line 0]
Stock insuffisant pour le produit 2 : demandé 10, disponible 2.

(1 row affected)
```

Essential server alerts

Specific alerts were created for errors **825** and **9002**, plus one alert for each severity from **19 to 25**. Each uses a **60-second delay between responses** to limit repeated notifications.

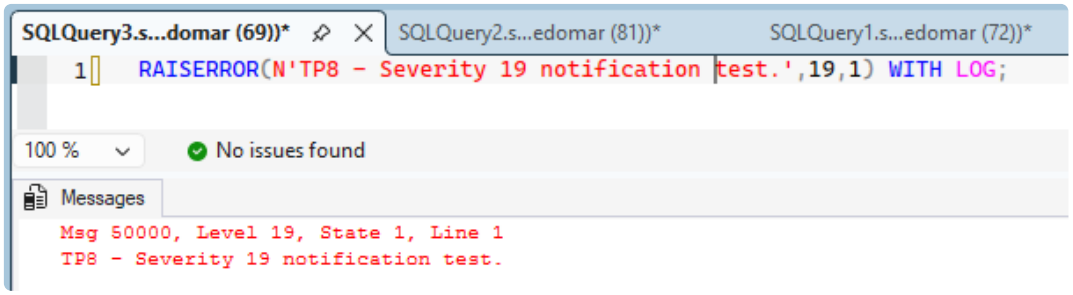
```
LQuery3.s...edomar (69))*   SQLQuery2.s...domar (81))*   SQLQuery1.s...edomar (72))*
7   while @s<=25
8   begin
9    set @name=N'Severity_'+cast(@s as nvarchar(2));
10   exec dbo.sp_add_alert @name=@name,@message_id=0,@severity=@s,@enabled=1,
11   @delay_between_responses=60,@include_event_description_in=1;
12   exec dbo.sp_add_notification @alert_name=@name,
13   @operator_name=N'AdminDBA',@notification_method=1;
14   set @s=@s+1;
15  end;
```

0 % No issues found

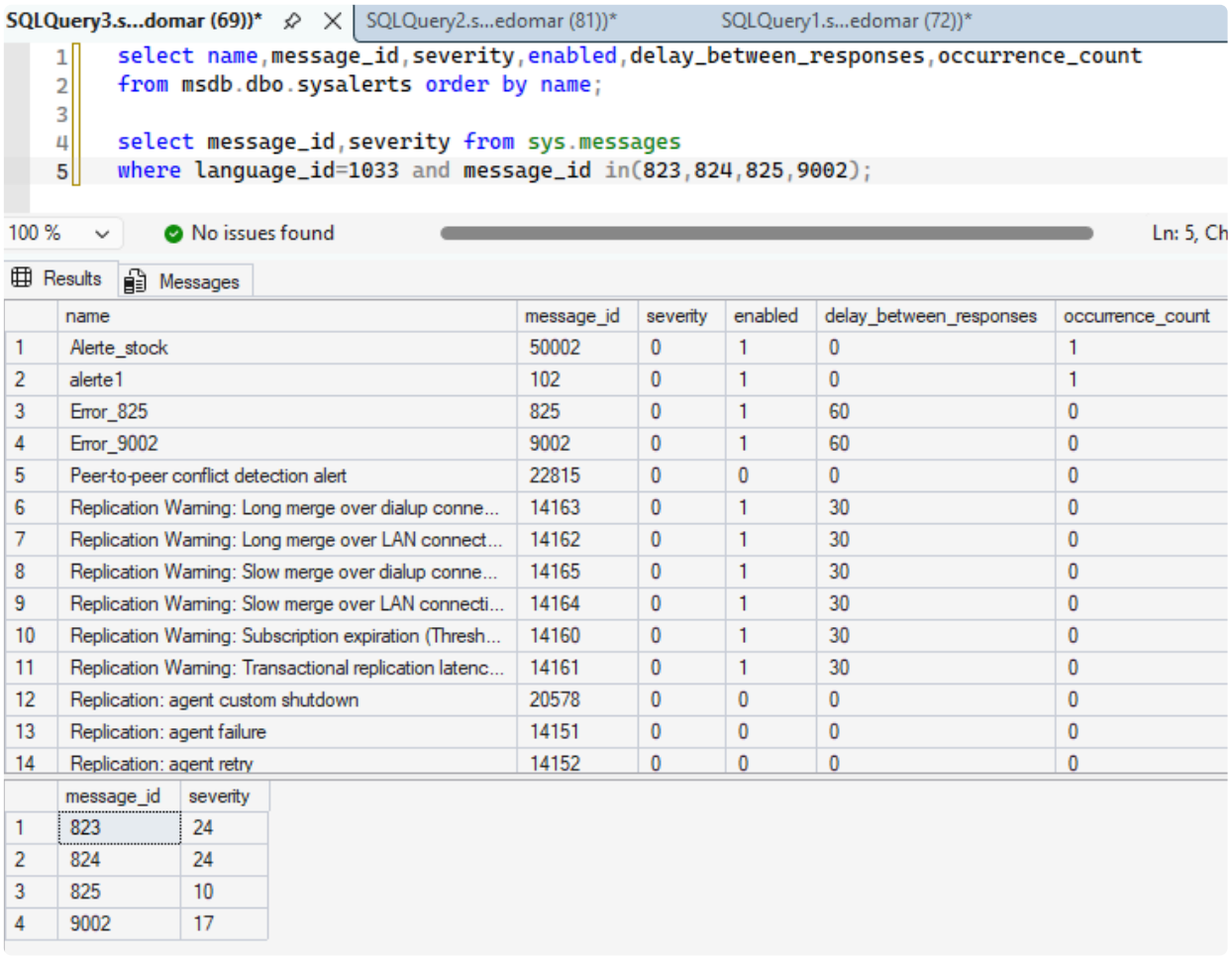
Messages

```
Commands completed successfully.
```

A simulated severity-19 event tested the notification without causing a real server fault.



The event generated the `Severity_19` email. The configuration query confirms the alerts, and `sys.messages` confirms the error severities.



	name	message_id	severity	enabled	delay_between_responses	occurrence_count
1	Alerte_stock	50002	0	1	0	1
2	alerte1	102	0	1	0	1
3	Error_825	825	0	1	60	0
4	Error_9002	9002	0	1	60	0
5	Peer-to-peer conflict detection alert	22815	0	0	0	0
6	Replication Warning: Long merge over dialup conne...	14163	0	1	30	0
7	Replication Warning: Long merge over LAN connect...	14162	0	1	30	0
8	Replication Warning: Slow merge over dialup conne...	14165	0	1	30	0
9	Replication Warning: Slow merge over LAN connecti...	14164	0	1	30	0
10	Replication Warning: Subscription expiration (Thresh...	14160	0	1	30	0
11	Replication Warning: Transactional replication latenc...	14161	0	1	30	0
12	Replication: agent custom shutdown	20578	0	0	0	0
13	Replication: agent failure	14151	0	0	0	0
14	Replication: agent retry	14152	0	0	0	0

	message_id	severity
1	823	24
2	824	24
3	825	10
4	9002	17

Error	Severity	Why it is monitored
823	24	Physical I/O error
824	24	Logical consistency error during I/O
825	10	Read succeeded only after retries
9002	17	Transaction log is full

A **severity-24 alert covers both 823 and 824** because both have that severity. Errors **825 and 9002 require specific alerts** because severities 10 and 17 fall outside the configured

19-25 range. A successful retry does not make error 825 harmless: it can indicate an underlying storage problem.